

Downscaling SMIPS soil moisture to 30 m: statistical and process-based estimation of root-zone soil moisture, with validation in the Murrumbidgee catchment

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Abstract

Root-zone soil moisture at field scale is a primary input to irrigation scheduling, yield forecasting and drought monitoring, yet operational products resolve kilometres rather than fields. We develop and validate two routes to a 30 m daily estimate of root-zone (0–90 cm) soil moisture for Australia, trained and evaluated against the OzNet network in the Murrumbidgee catchment (37 stations, 2006–2010, extended to 2002–2013). The first is statistical: a gradient-boosting downscaler of the ≈ 1 km TERN SMIPS **TotalBucket** product, drawing on national 30 m terrain (Copernicus DEM), ~ 90 m soil properties (SLGA), strictly backward-looking SMIPS aggregates, and antecedent SILO meteorology. The second is process-based: a calibrated single-layer bucket water balance forced only by SILO rainfall and potential evapotranspiration, with a ridge-regularised static offset derived from soil, terrain and climate covariates; it requires no SMIPS input and can be evaluated for any date. Skill is reported under four cross-validation designs of increasing strictness — leave-one-station-out, leave-one-block-out over nine spatially independent districts, leave-one-year-out, and block $\times$ year — because validation design alters the results more than any single change of model: for the process model the pooled Nash–Sutcliffe efficiency is +0.475 (year), +0.408 (station), +0.322 (block) and +0.273 (block $\times$ year). The statistical model is the stronger interpolator (blocked pooled NSE +0.36); the process model generalises better to unobserved districts (block-median +0.25 against +0.09); and the two fail in different places. Both approaches identify soil texture, independently, as the covariate that supplies between-site information. Decomposition of the residual error shows that the remaining deficit lies almost entirely in the site-level mean rather than the dynamics: the median held-out per-station correlation is 0.82, a median 52 % of mean-squared error is attributable to bias, and removing each station’s mean offset — an oracle bound — raises the median per-station NSE from +0.13 to +0.61. We further document two methodological defects that materially inflated interim results, a look-ahead leak in climatology features (≈ 0.13 NSE) and neighbour leakage in station-out validation, and a data-quality finding that limits the usable record: undocumented per-station calibration discontinuities restrict the 25-year OzNet archive to a homogeneous 2002–2013 era. The final process model comprises 14 interpretable parameters, uses only public national inputs, and produces 30 m fields for arbitrary dates. Within the sampled climate envelope it transfers to unobserved districts at pooled NSE $\approx +0.32$; beyond the envelope the error is confined to the site-level mean, reflecting the limits of the observation network rather than of either model.

Keywords: soil moisture; downscaling; SMIPS; machine learning; water balance model; spatial cross-validation; OzNet; Murrumbidgee

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1 Introduction

Daily root-zone soil moisture at the scale of individual fields would inform irrigation timing, trafficability, sowing decisions and drought response, yet existing operational products resolve 1–10 km. In Australia, the Soil Moisture Integration and Prediction System (SMIPS; CSIRO, 2021) provides a daily ≈ 1 km analysis of profile soil water (`TotalBucket`, mm); microwave products such as SMAP and ASCAT are coarser still, and sense only the near surface. The information needed to distribute moisture *within* a kilometre cell — terrain convergence, soil texture, aspect-dependent radiation load — is available nationally as static grids at 30–90 m resolution. This motivates a statistical downscaling: learn from in-situ profile observations how fine-scale terrain and soil redistribute the coarse-cell water, and apply the learned relation at every 30 m pixel.

This paper develops that approach and, in parallel, its physical counterpart. Two questions structure the work.

1. **How much of a held-out site’s root-zone series can national gridded covariates predict?** The downscaling is developed as a sequence of eight models: five statistical configurations culminating in a gradient-boosting model with antecedent meteorology (model 6), followed by a process-based alternative in which a five-parameter bucket water balance replaces the statistical estimator (model 7) and acquires a soil–terrain–climate readout (model 8).
2. **Which validation design yields an unbiased estimate of that skill?** The available stations are strongly clustered, and we show that the standard leave-one-station-out estimate largely measures *interpolation*. Every model is therefore evaluated under four hold-out designs of increasing strictness, and we argue that the strictest — block $\times$ year — is the estimate relevant to a national product.

Three findings extend beyond this catchment. First, validation design dominates model choice. For a fixed model, moving from year to station to block to block $\times$ year hold-outs lowers pooled skill monotonically ($+0.475 \rightarrow +0.273$ NSE for model 8), and the two approaches exchange rank between designs: the machine-learning model prevails under interpolation, the process model under spatial transfer. Comparisons of downscaling methods are therefore ambiguous unless the hold-out geometry is stated. Second, the error separates cleanly into level and dynamics. Both approaches reproduce temporal dynamics at held-out sites nearly as well as the data permit (median $r \approx 0.82$) and fail almost entirely through a constant per-site offset; in this data regime the limiting problem is not the temporal behaviour of soil moisture but the absolute level of an unvisited site. Third, two independent modelling approaches converge on the same covariate — soil texture — as the carrier of that level information, which is stronger evidence than either provides alone that the signal is physical.

We also document two corrections that altered interim conclusions: a look-ahead leak in pixel-climatology features that inflated cross-validated NSE by ≈ 0.13 and, once removed, reversed the apparent optimal regularisation regime; and logger sentinels together with per-station calibration discontinuities that render only 12 of 25 years of the in-situ archive usable. Both defects are liable to arise in comparable studies, and neither was detectable from the headline metrics.

The intended product is national. Every covariate employed is an Australia-wide public grid, so both models can be evaluated anywhere in the country; all development and validation to date, however, is confined to the Murrumbidgee, and we are explicit throughout about what that does and does not establish.

2 Data

2.1 Study area and in-situ reference: OzNet

The Murrumbidgee catchment ($\sim 82,000 \text{ km}^2$, New South Wales) spans a substantial climate gradient: across the stations used here, mean annual rainfall ranges from 318 to 1006 mm and the aridity index (mean P/PET) from 0.15 to 0.60. The OzNet monitoring network (Smith et al., 2012) comprises profile soil-moisture stations in three clustered sub-catchments — Yanco (13 stations, semi-arid plains), Kyeamba (13, undulating grazing country) and Adelong (wet, steep) — together with scattered regional sites M1–M7 across the catchment. After coordinate resolution and quality control, 37 stations enter the analysis; M3 lacks resolvable forcing and is excluded.

The regression target is the daily root-zone (0–90 cm) volumetric water content (%), computed as the mean of the 0–30, 30–60 and 60–90 cm sensor layers to match the SMIPS `TotalBucket` profile definition. Days missing any of the three layers are discarded rather than averaged from a partial profile, and the surface layer (0–5/0–8 cm), whose definition varies between file generations, is excluded. One station (Y3) integrates a single layer rather than three; it is retained, and the consequence is discussed in Section 6.3.

The primary study period is 2006–2010 (47,786 station-days), the window used for all model development. Section 5.5 extends the record to the homogeneous 2002–2013 era (101,524 station-days) identified by the quality-control analysis of Section 2.6.

2.2 Coarse predictor: SMIPS

SMIPS `TotalBucket` (CSIRO; $\approx 1 \text{ km}$, daily, mm of profile soil water) is the field to be downscaled, retrieved from the TERN GeoServer WCS as raw float32 values. One retrieval detail proved consequential for reproducibility. The WCS fits an integer pixel count to the requested bounding box, so the returned grid origin depends on the request window, and a fixed station can fall in different $\approx 1 \text{ km}$ cells for different windows: at one station the sampled value ranged from 43.7 to 61.6 mm across three request windows for the same date. Every request is therefore snapped to the native grid (origin 112.90499°E , -43.73500°N , cell size 0.0099976°), which renders sampled values independent of the request window (verified to 0.000 mm) and makes clustered multi-station retrievals exactly equivalent to per-station ones (Fig. 2).

2.3 Fine-scale static covariates: terrain and soil

Terrain covariates are derived at 30 m from the Copernicus DEM (European Space Agency, 2021): elevation, slope, aspect decomposed into northness and eastness, topographic wetness index (TWI), heat-load index (HLI) and flow accumulation. The DEM is reprojected to metric UTM coordinates before differentiation — gradients computed on geographic coordinates overstate slope by roughly three orders of magnitude — and each derivative is masked to the valid DEM footprint eroded by one pixel.

Soil covariates are taken from the Soil and Landscape Grid of Australia (SLGA v2, $\sim 90 \text{ m}$; Grundy et al., 2015): clay fraction, sand fraction, available water capacity (AWC) and bulk density, each depth-weighted over the standard slices spanning 0–100 cm to correspond with the target profile.

2.4 Meteorological forcing: SILO

SILO/DataDrill (Jeffrey et al., 2001) supplies daily rainfall, Morton potential evapotranspiration (Morton, 1983) and vapour-pressure deficit on a national $\approx 5 \text{ km}$ grid. The statistical model uses trailing aggregates as features (rainfall over the preceding 7, 30 and 365 days, $P - PET$ over 30 and 365 days, vapour-pressure deficit over 30 days, and the 365-day rainfall anomaly); the

process model uses the daily series directly as forcing. SILO is retrieved from one year before each study window so that every 365-day statistic is complete from the first day.

2.5 The training table

The statistical models consume a single long-format table with one row per station-day: the target; the coarse SMIPS value at the station’s grid-aligned pixel; seven terrain covariates; four soil covariates; four SMIPS lookback features (trailing 7-, 30- and 365-day means, and the departure of the current value from the trailing annual mean); seven antecedent-meteorology features; and cyclic day-of-year terms. Station coordinates are never used as predictors — they would permit a model to encode station identity — and the station label serves only to group cross-validation folds. Every temporal statistic is strictly backward-looking (Section 3.3).

2.6 Quality control and the usable record

Extending the archive beyond the development window exposed two data-quality findings that constrain all subsequent analysis.

Logger sentinels. The 2001–2025 archive contains 2,979 rows (1.85 %) outside the physical range, dominated by 65535 ($2^{16} - 1$) and other raw-count sentinels; these are concentrated after 2020 and absent from 2006–2010. They are removed, and nothing else is. Two candidate spike detectors — a robust- z rule against a 31-day rolling median, and a persistence test — were evaluated against independent rainfall and rejected: their detections were enriched in preceding rainfall by factors of 1.71 and 1.54 respectively, indicating that they preferentially removed genuine wetting events. Spike detection is retained only as a reported flag.

Calibration discontinuities. Trained on the full 25 years, model 8’s skill is positive through 2013 and collapses thereafter (2017–2019 pooled NSE -1.3 to -2.1 with -7 pp bias; negative correlation by 2023–2025). Comparison of the observations against rainfall identifies the cause: 18 of the 19 stations reporting in both eras read wetter in the later record without commensurate rainfall, and in 2019 — the Tinderbox Drought, among the driest years on record — the network reports the wettest mean soil moisture in the entire archive. A rain-adjusted step scan attributes the shift to per-station discontinuities (15 stations, in both directions, scattered in time), consistent with sensor replacement and recalibration. Such steps are invisible to row-wise filtering because every affected value is individually plausible. The homogeneous, usable era is 2002–2013; recovery of 2014–2025 would require per-segment recalibration rather than filtering.

3 Evaluation methodology

3.1 Metrics

For paired predictions and observations we report RMSE, bias, unbiased RMSE ($\text{ubRMSE} = \sqrt{\text{RMSE}^2 - \text{bias}^2}$; Entekhabi et al., 2010), Pearson r , and the Nash–Sutcliffe efficiency (NSE; Nash and Sutcliffe, 1970); $\text{NSE} > 0$ indicates greater skill than predicting the observed mean. Each set of out-of-fold predictions is scored in two ways: **pooled** across all station-days, a lenient view in which between-site variance enters the denominator and the separation of wet from dry sites is credited; and **per-station** within each station’s own series, the stricter within-site temporal test. The gap between the two is central to the study (Section 5.6), and both are reported throughout.

3.2 The validation ladder

All reported skill is out-of-fold under one of four hold-out designs of increasing strictness, implemented in a single harness so that models are compared on identical folds (Table 1).

Table 1: The four hold-out designs.

rung	design	withholds	measures
1	leave-one- year -out	the weather regime	a further year at instrumented sites
2	leave-one- station -out (LOSO)	one pixel	interpolation near instrumented sites
3	leave-one- block -out	a whole district (9 blocks)	transfer to unobserved districts
4	block $\times$ year	district <i>and</i> year (44 cells)	lower bound for a national product

The block design responds to a bias inherent in station-out validation on this network: 26 of the 37 stations lie in two tight clusters (Yanco’s 13 stations span 0.57° , Kyeamba’s 13 span 0.26°), so a held-out station’s neighbours share its ~ 5 km forcing cells and its soil map units and remain in the training set. The effective number of independent locations is therefore not 37 but nine: Yanco, Kyeamba, Adelong, and the six regional M-sites. Section 5.3 quantifies how much of the station-out result is attributable to this neighbour leakage.

In every design a fresh estimator is fitted per fold, so no fitted state crosses fold boundaries. Any training-side weighting is applied to training only; validation is always unweighted.

3.3 A look-ahead correction

An earlier version of this work computed SMIPS “climatology” features as each pixel’s full-period mean and standard deviation, allowing every training day to draw on the remainder of the record, including its own future. The leak inflated LOSO NSE by ≈ 0.13 and, more damagingly, made extreme regularisation (three-leaf trees) appear optimal, because small trees could not overfit the leaked level signal. All features in the present work are strictly trailing windows, seeded from data preceding the study window; estimators were re-tuned after the correction (GroupKFold on station, never on the reported score), whereupon the optimum reversed to large trees regularised by feature subsampling. We describe the defect because it was undetectable from any reported metric and was found only by audit of the feature construction; climatology features of this form are common in the downscaling literature.

4 Models

The two approaches share the target, the forcing data and the validation harness, and differ in everything else. Models 1–6 are statistical; models 7–8 are process-based.

4.1 Statistical models (1–6)

Model 1 — Random Forest baseline. The target is predicted from SMIPS, the seven terrain covariates and seasonality. On an initial four-station Kyeamba subset, which spans roughly one SMIPS cell, pooled LOSO skill is negative (NSE = -1.16): within-site correlation is high, but the coarse predictor carries no between-station signal. On 30 stations across the three catchments the model reaches pooled NSE = $+0.15$ ($r = 0.54$). Its characteristic failure, visible in every subsequent model, is shrinkage of predicted levels towards the training mean: dry stations are predicted too wet, wet stations too dry (Fig. 3).

Model 2 — regularised linear model. Standardisation followed by ridge regression on the same features. Pooled skill falls to NSE = $+0.02$ — a single global hyperplane extracts less of the between-site signal — but per-station tracking improves on the forest (12/30 against

7/30 stations at positive per-station NSE), and the coefficients are interpretable and physically plausible (slope strongly negative, SMIPS positive). Richer linear variants offer no improvement, and full quadratic expansions fail severely under LOSO (NSE $\approx -36,000$): polynomial terms extrapolate violently to unseen combinations of features.

Model 3 — gradient boosting at stock settings. Histogram gradient boosting (`HistGradientBoostingRegressor` Pedregosa et al., 2011) with default hyperparameters: pooled +0.12, between the linear model and the forest, with the best per-station uBRMSE of the three (3.5%). The pattern is consistent: dynamics are reproduced; the per-station level is biased.

Model 4 — SMIPS lookback and soil, with a tuned estimator. Two feature groups are added: strictly trailing SMIPS windows (7-, 30- and 365-day means, and the departure of the current value from the trailing annual mean), which give the model a level–departure decomposition of the coarse signal; and the four SLGA soil covariates. Soil alone degrades models 1–3 under LOSO — sand fraction acts as a nearly unique station identifier — yet, conditioned on the lookback level anchor, the same covariates help. The contribution of a static covariate thus depends on the information already present. On the 30 dense-cluster stations model 4 marginally exceeds the baseline (+0.17 against +0.15); the features become effective only once the network gains spatial extent. With the six regional M-sites added (36 stations), model 1 turns negative (−0.05) while model 4 reaches +0.31 (Fig. 4). A leave-region-out check — train on two catchments, predict the third, a split never used in model selection — confirms that the gains generalise: +0.12 against −0.72 for the baseline.

Model 5 — soil smoothing, a quantified trade-off. Gaussian smoothing of the soil rasters removes the blocky SLGA map-unit boundaries from the 30 m product and improves a single-date comparison against stations, but degrades cross-validated skill monotonically with blur radius; even $\sigma = 1$ px costs ≈ 0.07 NSE. The sharp soil detail that disfigures the map is the same signal that discriminates between-station levels; the two cannot be separated, and model 5 is not recommended.

Model 6 — antecedent meteorology (the preferred statistical model). The seven SILO trailing-window features are added to model 4. Tuned without leakage, this feature set selects the opposite regularisation regime from model 4: large trees (127 leaf nodes) with strong per-split feature subsampling (15 % of features per split). Permutation importance is led by soil texture (sand, clay), followed by terrain (slope, elevation), the SMIPS lookback and the annual water balance (Fig. 6). The prominence of soil warrants caution as well as confidence: it is also the covariate best able to act as a station fingerprint.

4.2 Process-based models (7–8)

Model 7 — calibrated bucket water balance. The statistical estimator is replaced by simulation: a single-layer daily water balance over the 0–90 cm profile, forced only by SILO rainfall P and Morton potential evapotranspiration PET,

$$S_{t+1} = \text{clip}(S_t + P_t - \text{AET}_t - k S_t, 0, s_{\max}), \quad \text{AET}_t = \text{PET}_t \cdot \min\left(1, \frac{S_t}{\alpha s_{\max}}\right), \quad (1)$$

$$\text{VWC}_t [\%] = \theta_r + \Delta\theta \cdot \frac{S_t}{s_{\max}}, \quad (2)$$

with five global parameters ($s_{\max}$, α , k , θ_r , $\Delta\theta$) calibrated on the training stations within each fold — the process-model analogue of estimator fitting — so that model 7 passes through the same validation harness as models 1–6. Simulation begins a full year before any scored date, at

half capacity, to eliminate sensitivity to the initial condition. The model uses no SMIPS, no soil, no terrain and no machine learning.

A two-stage ridge offset supplies the single degree of freedom the bucket structure offers for between-site variation. After calibration, the per-station mean residuals (one sample per training station) are ridge-regressed on standardised terrain statics (TWI, slope, elevation), with the penalty chosen by leave-one-out over the training stations; a held-out station receives its offset from its own covariates, never from its observations. Two negative results shaped this stage. Unregularised least squares lowers held-out skill — with ≈ 36 station-level samples the coefficients fit the training stations rather than the relation, and the selected ridge penalty is correspondingly heavy ($\alpha_{\text{ridge}} \approx 46$). Climate normals in the offset reduce station-out skill — a result that reverses under blocked validation (Section 5.3) and illustrates directly how the validation design conditions the conclusion.

Model 8 — the bucket with a full static stack (the preferred process model). Model 7 with four additions, each evaluated separately under both station-out and blocked designs:

1. **SLGA soil in the offset stage** (clay, sand, AWC, bulk density) — the change that first brought the process model to parity with the statistical one;
2. **an aridity static** (mean P/PET from the training-window forcing) — the level information that transfers across climates;
3. **SLGA AWC as per-station bucket capacity** — higher-capacity soils store and read out more water (capacity scales with AWC relative to its training mean);
4. **stratified training weights** — hierarchical over aridity tertile, then block, then samples, tempered as $\sqrt{w}$ — so that ten clustered Yanco stations no longer dominate three scattered dry M-sites.

The complete model comprises 14 interpretable parameters. The fitted offset coefficients are physically consistent in sign and magnitude, which supports their credibility at unobserved sites: clay +1.86 % per training standard deviation (finer soils hold more water), sand −1.77, TWI +0.69 (convergent terrain is wetter), slope −0.58, elevation −0.35, aridity +0.61 (wetter climate, wetter site).

Because model 8 carries a state rather than lookback features, it runs forward from a spin-up to any date without SMIPS. The fitted recession ($k = 0.0065 \text{ d}^{-1}$, 154-day e -folding time) implies, and direct measurement confirms, that any spin-up of six months or longer yields predictions identical to a ten-year spin-up; the released tool uses ≈ 2.4 years. For 30 m map inference the water balance runs per $\approx 5 \text{ km}$ SILO cell — the scale at which the weather itself varies — with the 30 m structure supplied by the per-pixel soil, terrain and aridity offsets (Fig. 13).

5 Results

5.1 Statistical models under station-out validation

Leave-one-station-out skill on the 36-station, 2006–2010 table is summarised in Table 2.

Model 6 exceeds model 4 on every metric (Fig. 5), and the progression of features is monotone and interpretable: the lookback windows supply a level–departure decomposition, soil supplies between-site texture once a level anchor exists, and the antecedent meteorology supplies the recent water balance. The median held-out per-station correlation is 0.81 — the dynamics are essentially resolved — while the median per-station NSE remains negative: the residual is a constant per site. A more conservative grouped cross-validation, holding out ~ 7 stations at once, gives $\approx +0.25$, an early indication of the leakage quantified in Section 5.3.

Table 2: Statistical models, leave-one-station-out, 36 stations, 2006–2010.

LOSO, 36 stations	model 1 (RF)	model 4	model 6
Pooled NSE / r	−0.05 / —	+0.31 / 0.58	+0.38 / 0.62
Median per-station NSE	−1.01	−0.57	−0.19
Median per-station bias	4.84 %	4.88 %	3.85 %
Stations with NSE > 0	—	10/36	16/36

5.2 Process models under station-out validation

Table 3: Process models, leave-one-station-out, 37 stations, 2006–2010. The model 6 column is the published 36-station reference, not a same-rows comparison.

37 stations, LOSO	model 7 (bucket + terrain)	model 8 (final)	model 6 (ref.)
Pooled NSE / r	+0.18 / 0.43	+0.41 / 0.64	+0.38 / 0.62
Median per-station NSE	−0.03	+0.13	−0.19
Median per-station r	0.83	0.83	0.81
Stations with NSE > 0	17/37	—	16/36

Two results stand out (Table 3). First, the five-parameter bucket already matches the 25-feature boosting model in within-station temporal skill (median r 0.83 against 0.81; median per-station NSE −0.03 against −0.19): rainfall minus evapotranspiration, routed through a single store, captures most of the year-to-year signal at a point (Fig. 7). What globally calibrated parameters cannot express is the between-site level — the predicted range compresses towards the middle while observations span 3–52 %, the same shrinkage observed for model 1, reached from the opposite direction. Second, soil is what resolves the level: adding SLGA to the offset stage raises the pooled score from +0.18 to +0.41, parity with the statistical model, and the same covariate family dominates model 6’s feature importance. That two independent methods identify soil texture as the carrier of between-site level information is strong evidence that the signal is physical rather than an artefact of either method.

Where models 7–8 err, they err by a constant: stations track the observed series at a fixed offset, with $r \geq 0.7$ at 30 of the 37 stations (Fig. 8). Quantifying this failure occupies the remainder of the paper.

5.3 Blocked validation: interpolation against transfer

Holding out the nine spatially independent blocks (Fig. 9) reproduces the deployment condition of a national product. Table 4 summarises the principal movements.

Table 4: Station-out against blocked skill, 2006–2010 (pooled NSE / per-station-median NSE).

¹Published 36-station reference.

configuration	station-out	blocked
model 8, soil + terrain only	+0.397 / +0.22	+0.222 / −0.18
+ aridity	+0.388 / +0.13	+0.256 / −0.02
+ aridity + weights	+0.403 / +0.13	+0.290 / −0.01
+ capacity + aridity + weights (final)	+0.408 / +0.13	+0.322 / +0.07
model 6	+0.377 ¹	+0.355 / −0.21
model 6 + weights	—	+0.379 / −0.27

Station-out results are substantially neighbour leakage. The pre-stack model 8 falls from +0.40 to +0.22 pooled under blocking, and its per-station median changes sign (+0.22 $\rightarrow$ -0.18): most of its apparent between-site level skill derived from same-cluster neighbours in the training set. Model 6’s pooled score barely moves (+0.38 $\rightarrow$ +0.36), but only because 76 % of the pooled rows belong to Yanco and Kyeamba and each large cluster compensates for the other; its block-median is +0.09.

The full stack is justified only under blocking. Aridity normals reduce station-out skill, replicating model 7’s negative result, yet contribute +0.03–0.07 pooled under blocking. AWC capacity is negligible in isolation — AWC varies by only $10.9 \pm 0.7\%$ across these stations — but combines effectively with the other components. The stratified weights improve model 8 monotonically and degrade model 6 (block-median +0.09 $\rightarrow$ -0.14): up-weighting individual stations drives a 127-leaf boosting model to fit their idiosyncrasies in a way a 14-parameter ridge cannot. Whether a covariate belongs in a model is thus not a fixed property: it depends on which deployment the validation represents. The final model 8 uses the full stack: blocked pooled +0.322, the first positive blocked station-median (+0.07), and a median $|\text{bias}|$ of 3.17 %.

Table 5: Per held-out block (dry $\rightarrow$ wet): NSE / bias (%).

block	aridity	model 6	model 8 (final)
M6	0.15	+ 0.59 / -0.8	+0.27 / -2.2
M7	0.16	-3.32 / +3.6	-4.38 / +4.5
M5	0.17	+0.14 / +1.3	+ 0.73 / +0.5
Yanco	0.18	+0.19 / +3.1	+0.17 / +1.5
M4	0.21	-0.19 / +3.5	+ 0.61 / +1.0
M1	0.31	+0.09 / +1.9	+0.13 / +2.3
Kyeamba	0.32	+0.41 / -1.2	+0.41 / -1.7
M2	0.33	-3.39 / +6.7	+ 0.74 / +0.2
Adelong	0.51	-0.33 / -4.5	-1.53 / -8.0

The two approaches fail in different places. Per-block results (Table 5; Fig. 10) show model 6 collapsing at M2 where model 8 performs well, and model 6 nearly surviving Adelong — its antecedent-rainfall features carry precisely the climate signal model 8’s statics lacked — where model 8 fails; both fail at M7. Failure concentrates at the edges of the climate envelope — the driest station, and the wettest cluster, whose aridity of 0.45–0.60 lies beyond the training maximum of 0.38 once it is withheld — and consists purely of level error: r remains between 0.65 and 0.91 at every failed block. Within the sampled envelope, transfer is respectable (+0.61 to +0.74 at M2, M4 and M5). No reweighting or covariate can compensate for an unpopulated region of predictor space; only observations there can.

5.4 Temporal validation and the strict design

Holding out years rather than places (Fig. 11) completes the ladder. Table 6 gives the full progression for the final model 8.

Temporal generalisation is markedly easier than spatial. The period 2006–2010 spans a $3.8\times$ range in annual rainfall (252–951 mm) across the tail of the Millennium Drought and its break, yet four of the five held-out years score within 0.05 NSE of one another. Only 2010 — wetter than anything in calibration — degrades, and in the same direction as the spatial wet edge: predicted too dry. Contrary to our expectation that a state-carrying process model would withstand a regime shift better than a feature-carrying statistical model, model 6 wins the year

Table 6: The validation ladder for the final model 8, 2006–2010.

design	withholds	pooled NSE	station-median NSE
leave-one-year-out	the regime	+0.475	+0.16
leave-one-station-out	one pixel	+0.408	+0.13
leave-one-block-out	the district	+0.322	+0.07
block $\times$ year	both	+0.273	−0.01

folds decisively (+0.771 against +0.475). That figure, however, describes instrumented sites: with no site withheld, 25 features suffice to identify each station and learn its level directly. The same model 6 records a block-median of +0.121 under block $\times$ year against the process model’s +0.332. The two designs reverse the ranking of the two approaches, which is precisely why the hold-out geometry must accompany every reported figure.

Under the strictest design the typical station’s NSE is zero: at a new site in an uncalibrated regime, the model outperforms a site-mean prediction only in the pooled sense (+0.273), by ranking wet districts against dry ones. The strict design also reveals a feature the single-axis folds concealed: the dry extreme (2006) is unproblematic when the site is known (+0.440) and nearly as difficult as the wet extreme when it is not (+0.146 under block $\times$ year). Knowledge of the site’s level renders unfamiliar weather manageable; without it, both extremes are comparably hard. One caveat applies: neither approach isolates time perfectly — lookback features and the continuously simulated bucket state both cross the fold boundary — so the year folds measure a year whose weather was absent from calibration, not a year in isolation.

5.5 Extending the record: 2002–2013

With sentinels removed and the homogeneous era identified (Section 2.6), model 8 was refitted on 2002–2013: twice the data, and twelve years spanning the Millennium Drought and the 2010–12 La Niña (Table 7).

Table 7: Original fit against extended-era refit (pooled NSE).

harness	2006–2010 fit	2002–2013 refit
station-out	+0.408	+0.431
blocked	+0.322	+0.316
block $\times$ year	+0.273	+0.304
blocked $\rightarrow$ block $\times$ year degradation	0.049	0.012

The gain grows with the strictness of the test, and the most informative statistic is the degradation from blocked to block $\times$ year folds: the refit loses only 0.012 pooled NSE when the test year is withheld in addition to the district, against 0.049 for the five-year fit. Training across regimes confers robustness to unseen regimes — the property a national product requires. A like-for-like comparison on the 50,608 station-days predicted by both fits clarifies the trade-off: the refit is better pooled (+0.426 against +0.410) and worse at the median station (+0.042 against +0.130), exchanging in-period per-site precision for calibration that holds across regimes. The refitted parameters shift coherently for a record containing more drought — a smaller ($s_{\max}$ 179 $\rightarrow$ 158 mm), slower-draining (k reduced by 22 %) bucket with a narrower readout range — with the same 14 parameters and no structural change.

5.6 Decomposition of the residual error

Separating each station’s held-out MSE into level and dynamics, $\text{MSE} = \text{bias}^2 + \text{ubRMSE}^2$ (Fig. 12), localises the remaining deficit. The median share of MSE attributable to level alone

is 52%; at 10 of 37 stations level accounts for more than 80% of MSE; the median per-station correlation is 0.82; and 16 stations combine $r > 0.5$ with $\text{NSE} < 0$.

The stations with the worst NSE include those with the smallest absolute errors in the network — one records the best ubRMSE of all 37 (0.91 pp) at $\text{NSE} = -2.1$. NSE charges the full cost of a constant offset, and at low-variance sites a small offset dominates the score. Removing each station’s mean offset — an oracle computed from held-out truth, hence an upper bound rather than an attainable result — raises the median per-station NSE from +0.13 to +0.61 and leaves 36 of 37 stations positive. Nearly all of the attainable improvement therefore resides in a single term: the absolute level of an unvisited site. Each successive model-8 configuration recovered a few hundredths of NSE against this term; the decomposition explains why the gains were small, and where a substantial one would have to originate (Section 6.4).

5.7 The 30 m product

Figure 1 shows the coarse ≈ 1 km SMIPS field beside the generated 30 m field through a seasonal cycle: within each coarse cell the model resolves drainage lines, wet valley floors and dry ridges, while the seasonal signal is carried by the coarse driver and the antecedent forcing. The process model produces fields of the same character for arbitrary dates without SMIPS (Fig. 13 shows a date fourteen years after its training period); its 30 m structure derives from the per-pixel soil, terrain and aridity offsets applied to a water balance run per ≈ 5 km forcing cell. In-situ observations are never an input to either model; they serve only as the training label and the validation reference. These fields are qualitative products, and the quantitative claims rest on the cross-validated tables above. A conventional mass-conservation constraint — each cell aggregating to the coarse observation — is deliberately absent: SMIPS is expressed in mm and enters as an input, not as an observed coarse reference in the target units. Section 6.4 outlines the rebasing this suggests.

6 Discussion

6.1 Reported skill is a property of the validation design

The largest single determinant of reported skill in this study is not the model but the hold-out geometry: +0.475 to +0.273 pooled for one fixed model, with the model rankings reversing between runs (model 6 leads under year folds and pooled blocks; model 8 leads on block-medians and under the strict design). Station-out validation — the community default — sits near the top of that range because of cluster-neighbour leakage that no per-fold hygiene can remove; only a change of fold geometry can. We suggest that downscaling studies on clustered networks report a block-level hold-out alongside station-out results as a minimum, and that claims of validity for arbitrary dates be tested with year and block $\times$ year folds.

6.2 Complementarity of the two approaches

The approaches are equal under interpolation, and each dominates a different form of generalisation: the boosting model’s antecedent features carry climate information into wet districts that the process model’s statics cannot reach, while the bucket’s physical structure transfers the level relation to districts for which the statistical model has no support. They fail at different blocks. The natural synthesis — bucket storage as a feature of the statistical model, or SMIPS assimilated into the bucket state — is the obvious next experiment (pursued beyond the scope of this paper), and the per-block complementarity indicates that it addresses a genuine gap rather than combining two copies of the same error.

6.3 Limitations

Geographic scope. All training and validation is confined to the Murrumbidgee. The covariates are national and both models can be evaluated anywhere, but output beyond the catchment carries an uncorrected per-site level bias and no validation; the released tools state this on every run. The effective independent sample is nine locations spanning aridity 0.15–0.60, and skill degrades to level failure at the edges of that envelope.

Measurement heterogeneity. The target is not perfectly depth-homogeneous: one station’s root zone is a single sensor layer, and its bias carries the sign a shallow reference would produce (+4.3%). With $n = 1$ this remains a hypothesis, but it raises the possibility that part of the between-site level error is a measurement inconsistency rather than a modelling failure, and it sets the reconciliation standard for any additional network. The archive’s calibration discontinuities (Section 2.6) limit the usable record to 2002–2013 and constitute a finding in their own right: long in-situ records cannot be assumed homogeneous, and row-wise quality control cannot detect step changes whose every value is individually plausible.

Model scope. The process model is a single-layer bucket with a linear readout; it cannot represent within-profile dynamics, lateral redistribution or irrigation. The statistical model can in principle absorb such effects, but has no mechanism for extrapolating beyond its training envelope, as the Adelong result demonstrates.

6.4 Priorities for further improvement

The error decomposition makes the priority unambiguous: further refinement of the temporal dynamics would be misdirected, because the binding constraint is the absolute level of unvisited sites — within the sampled envelope an information problem rather than a modelling one. Three routes follow. (i) **A national in-situ sample:** additional independent locations, not additional years at existing ones; observations from a single catchment cannot validate a continental product. (ii) **Rebasing on an observed coarse reference:** decompose the 30 m prediction into a cell mean and a fine anomaly, and rebase the mean onto an observed coarse soil-moisture field in the target units (for example SMAP L4 or ASCAT root-zone products). This addresses the level term directly with satellite rather than in-situ information, and is in our assessment the highest-value untested experiment. (iii) **Correction rather than removal** of the 2014–2025 record, via per-segment offsets at detected calibration breaks; this fits the ridge-offset architecture, but must remain a training-time nuisance parameter, since the static covariates do not change at a break.

7 Conclusions

We have constructed and validated two routes to a 30 m daily root-zone soil-moisture product from national public data: a gradient-boosting downscaler of the ≈ 1 km SMIPS product (model 6), and a calibrated bucket water balance with a static soil–terrain–climate readout that requires no SMIPS input (model 8). Under the conditions most favourable to each, both perform well: model 6 interpolates at pooled $\text{NSE} \approx +0.38$ near instrumented sites and reconstructs held-out years at +0.77; model 8 transfers to held-out districts at pooled +0.32 with 14 interpretable parameters, runs for any date, and requires a spin-up of no more than six months. Under the conditions the intended national product will face — a new place in an uncalibrated regime — the pooled NSE is $\approx +0.27$ –0.30, the typical station’s NSE is zero, and the failures concentrate at the edges of the climate envelope as level error alone, with correlation intact throughout (median r 0.82).

The methodological conclusions carry beyond the numbers. Validation geometry dominates model choice, reverses model rankings, and determines which covariates appear beneficial.

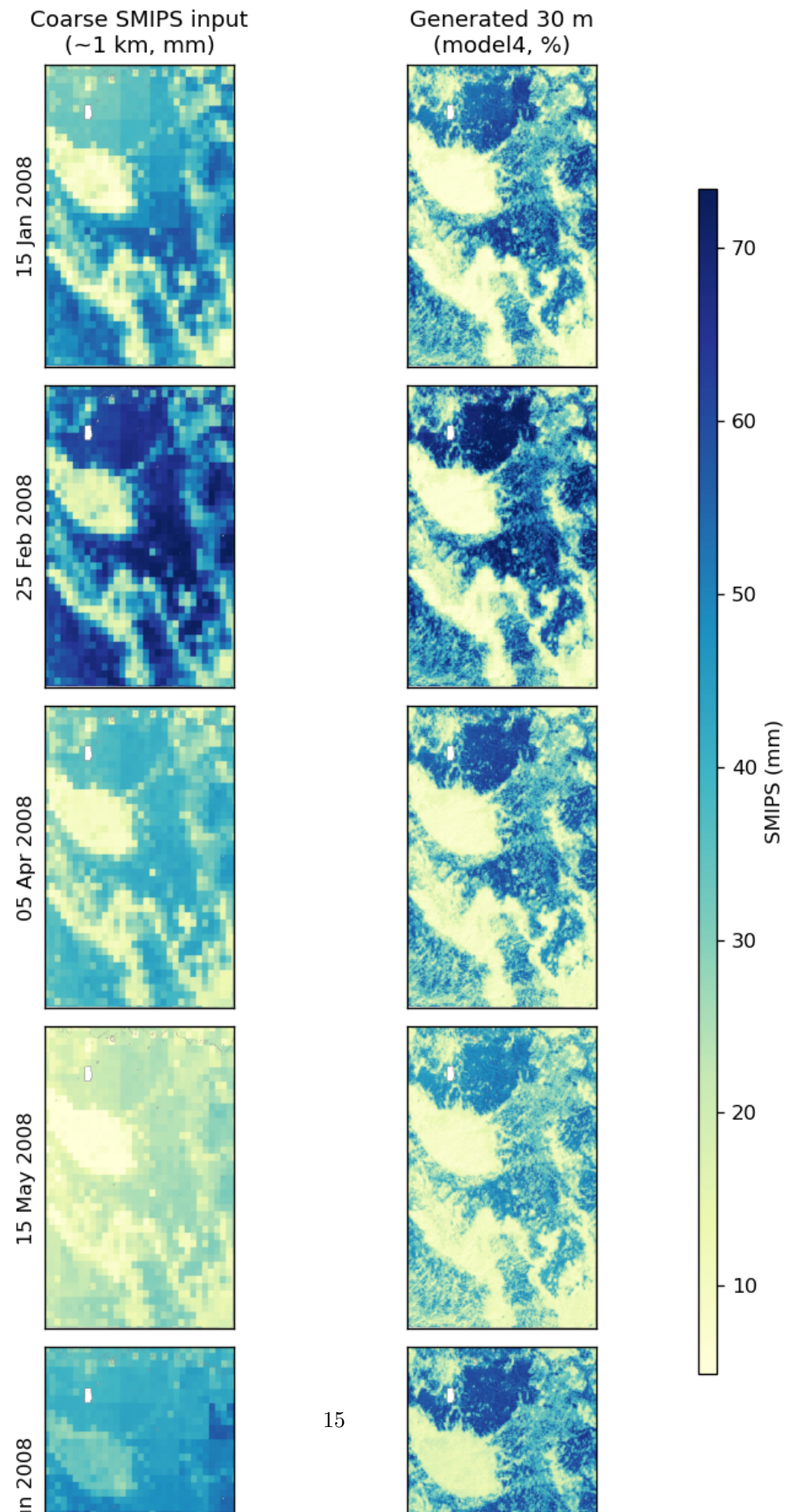
Look-ahead leakage in climatology features inflates skill and corrupts hyperparameter selection simultaneously. Clustered networks reduce station-out validation to an interpolation estimate. And a 25-year in-situ archive was limited to twelve usable years not by outliers but by undocumented per-station recalibration, exposed only through comparison with an independent physical reference. The remaining deficit — the absolute level of sites without sensors — is now precisely localised, and the most promising next step is to anchor predicted cell means to observed coarse satellite soil moisture rather than to refine either model further.

Code and data availability

The implementation, trained artefacts (`model6.joblib`, `model8.joblib`), the validation harness, and scripts regenerating every table and figure are available in the project repository (`emt/` package; `handout/` documentation). All inputs are public national products: SMIPS and SLGA via TERN, SILO/DataDrill forcing, the Copernicus DEM, and the OzNet archive (<https://www.oznet.org.au>). SLGA requires a free TERN API key and SILO a registered email address.

Figures

Coarse SMIPS → generated 30 m, Kyeamba 2008



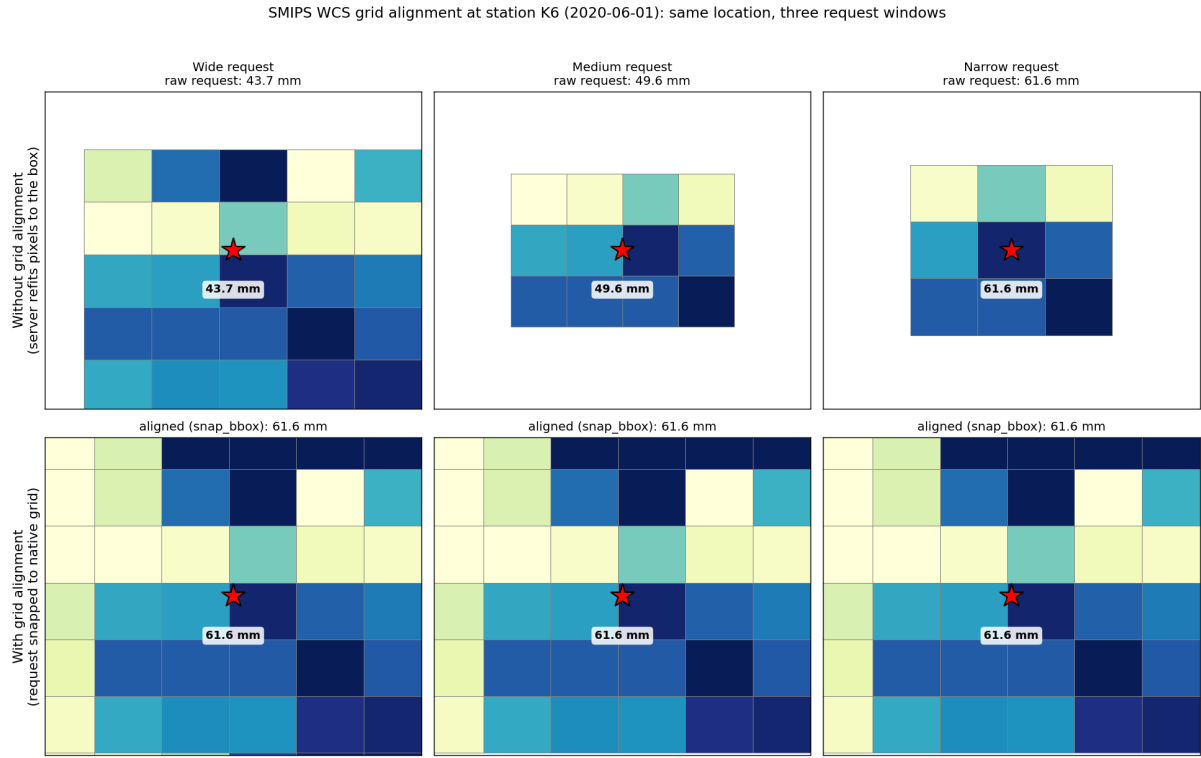


Figure 2: SMIPS retrieval: snapping requests to the native WCS grid removes the dependence of sampled values on the request window.

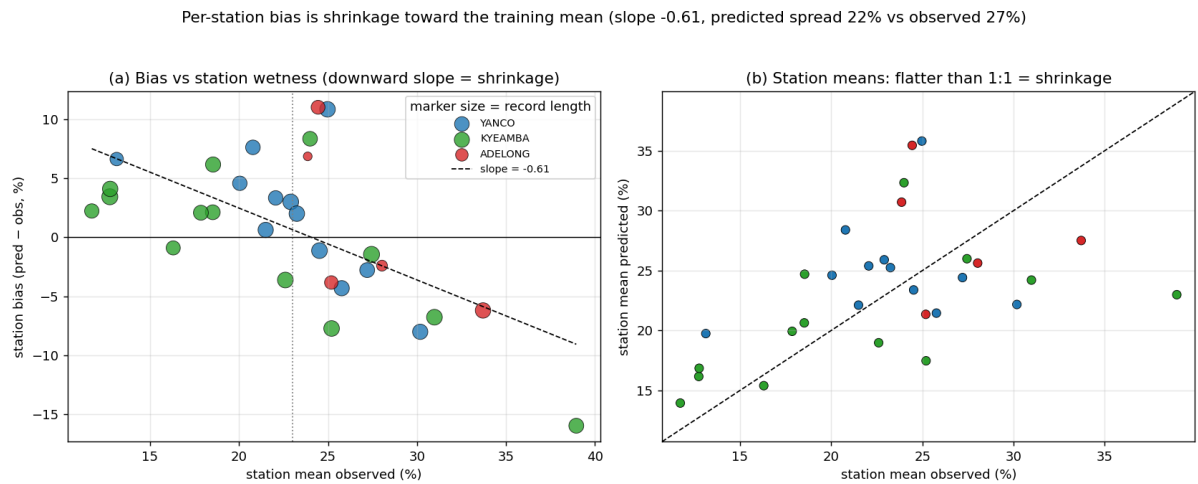


Figure 3: The baseline failure mode (model 1): per-station bias is shrinkage towards the training mean — dry stations predicted too wet, wet stations too dry.

Extending coverage: +6 regional Murrumbidgee sites (30 → 36 stations)

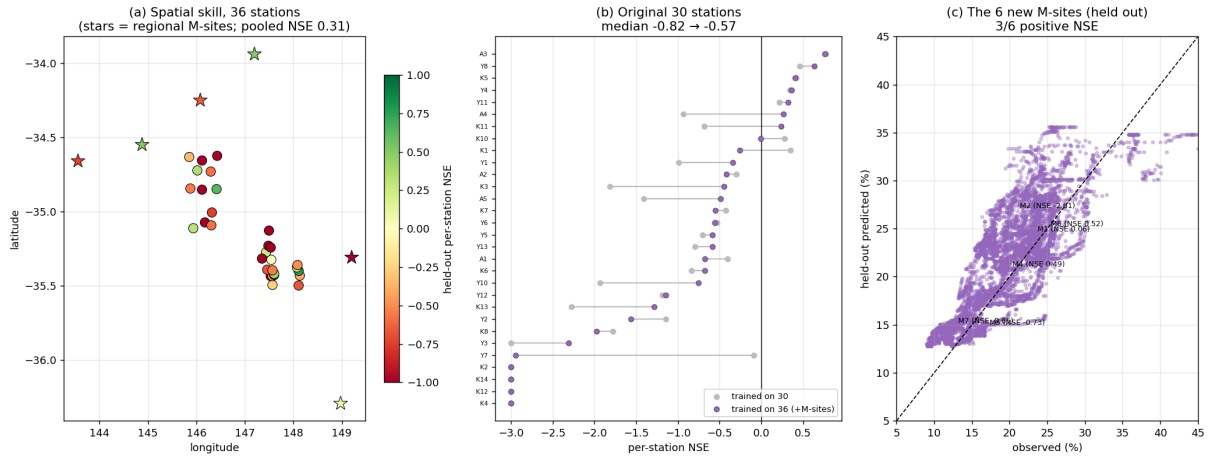


Figure 4: The effect of spatial coverage: extending from 30 to 36 stations with the regional M-sites drives model 1 negative (-0.05) while model 4 reaches $+0.31$.

model6 (honest, leak-free), leave-site-out over 36 stations, 2006-2010

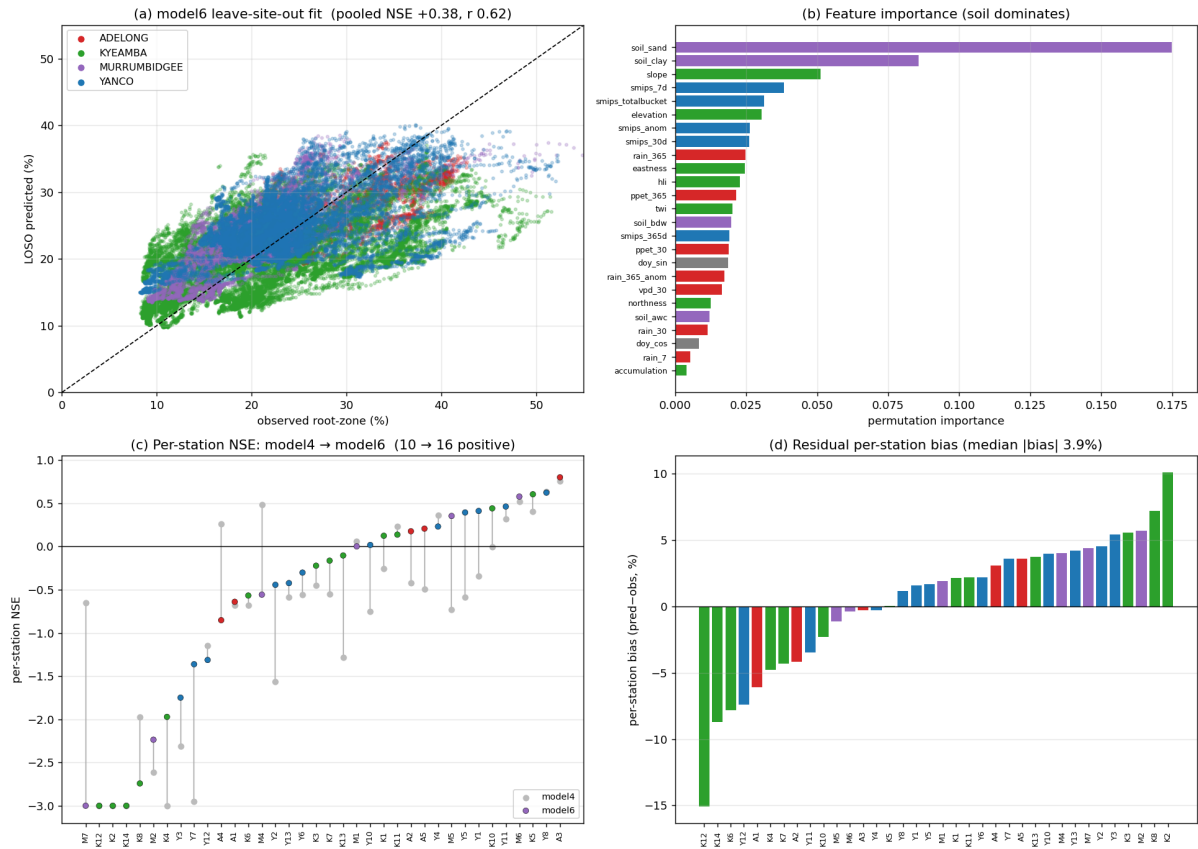


Figure 5: Model 6 leave-station-out results: fit, feature importance, model 4 $\rightarrow$ 6 per-station NSE, and residual bias.

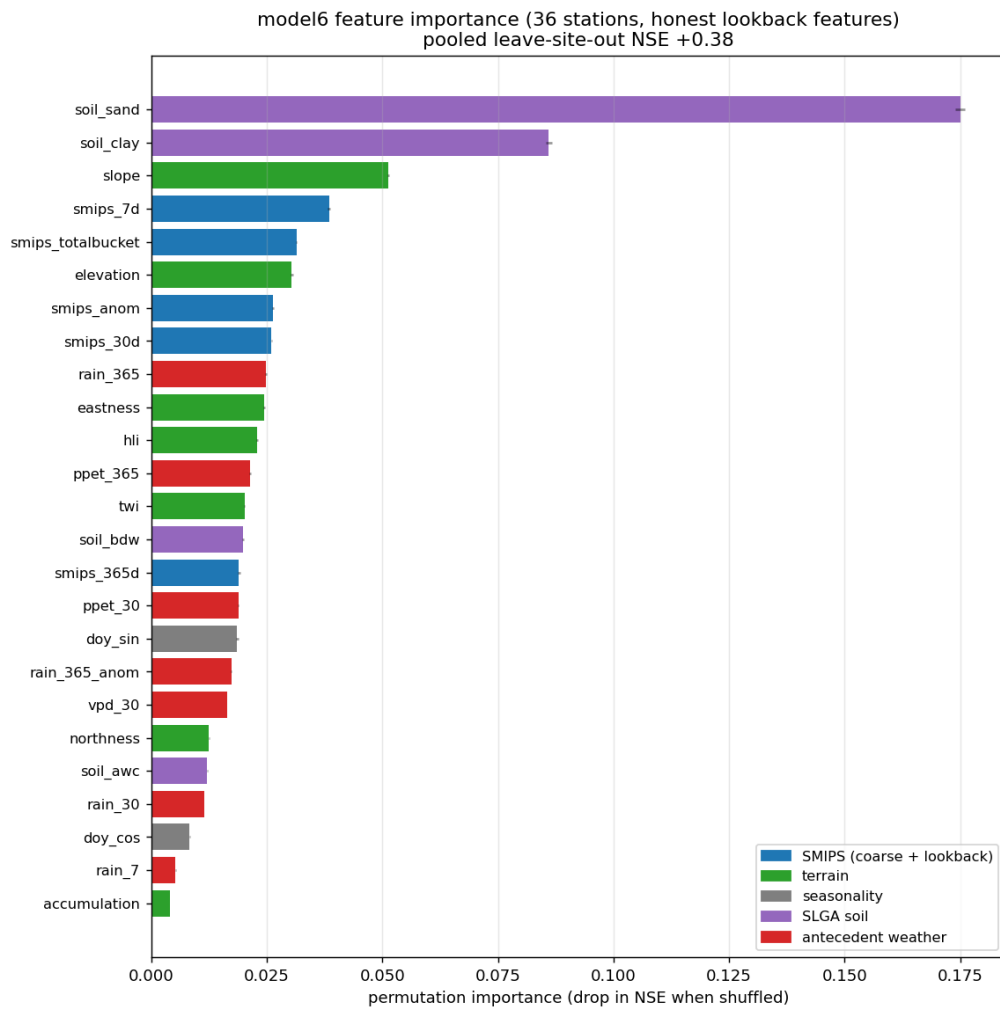


Figure 6: Model 6 permutation importance on the leak-free features: soil texture leads, followed by terrain, the SMIPS lookback and the annual water balance.

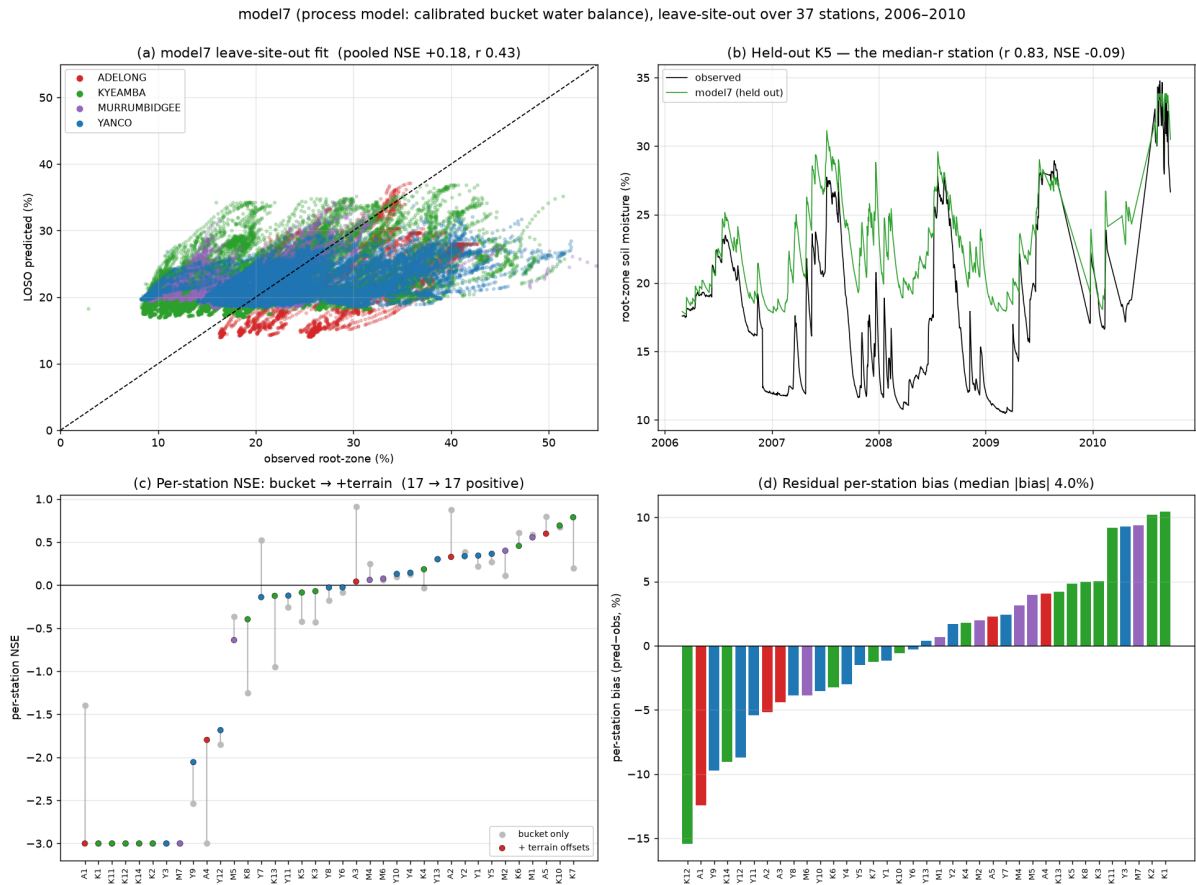


Figure 7: Model 7 (bucket water balance) leave-station-out results: the predicted range compresses towards the middle (level shrinkage) while a held-out station's dynamics are tracked through five years.

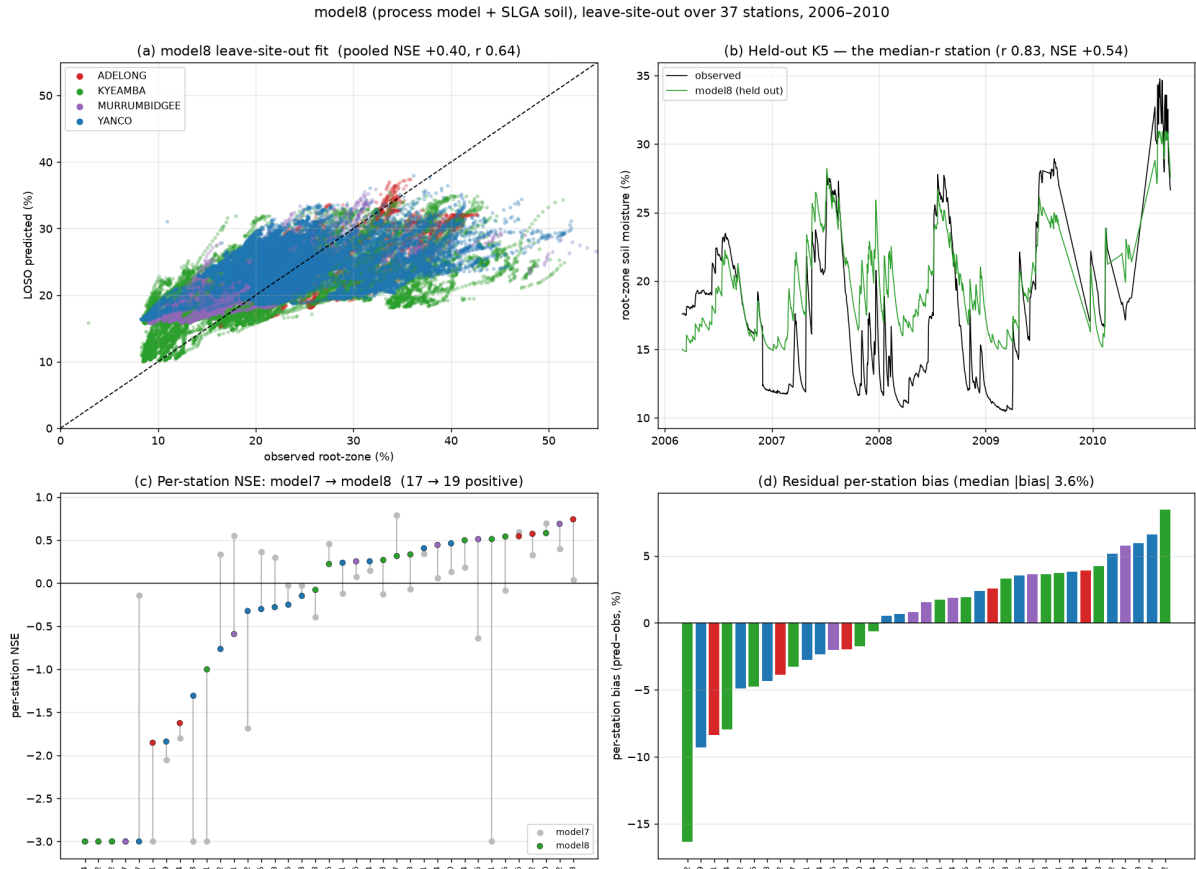


Figure 8: Model 8 leave-station-out results (pre-stack reference configuration).

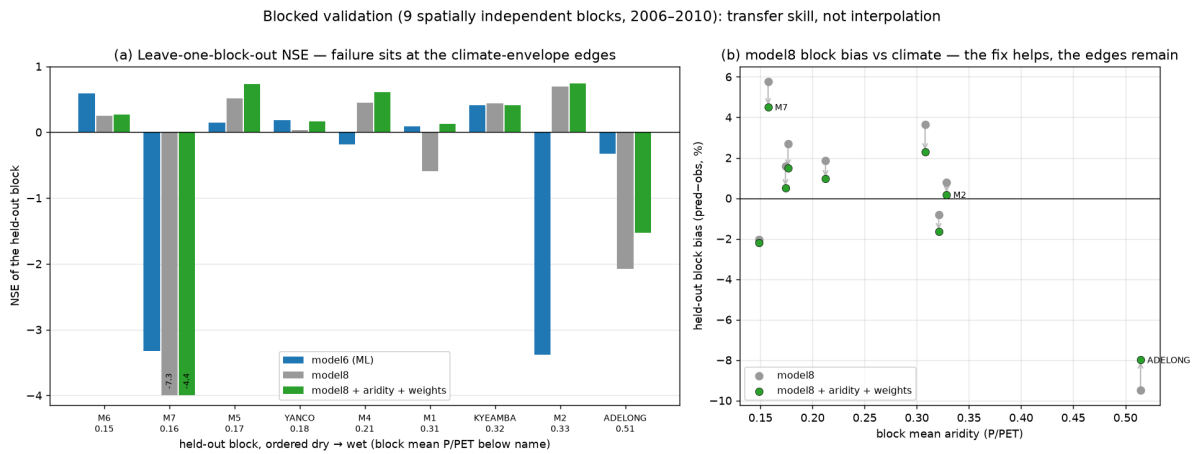


Figure 9: Blocked validation: the configuration ladder for models 8 and 6 under leave-one-block-out over nine spatially independent districts.

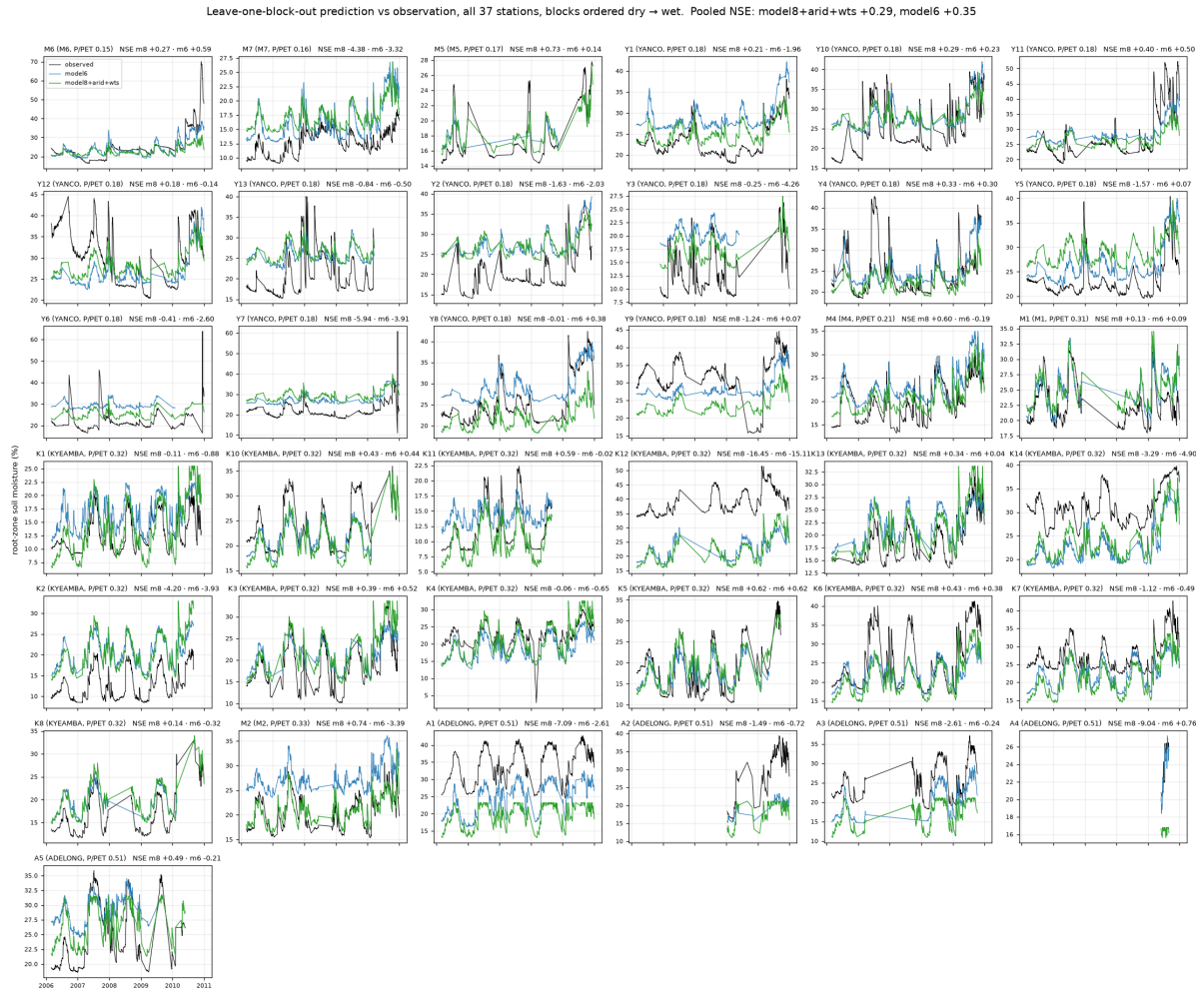


Figure 10: Blocked hold-out, per-station series (observed in black; model 6 blue; final model 8 green), grouped by block from dry to wet. Both models track the wetting–drying cycle; the errors manifest as whole-series offsets.

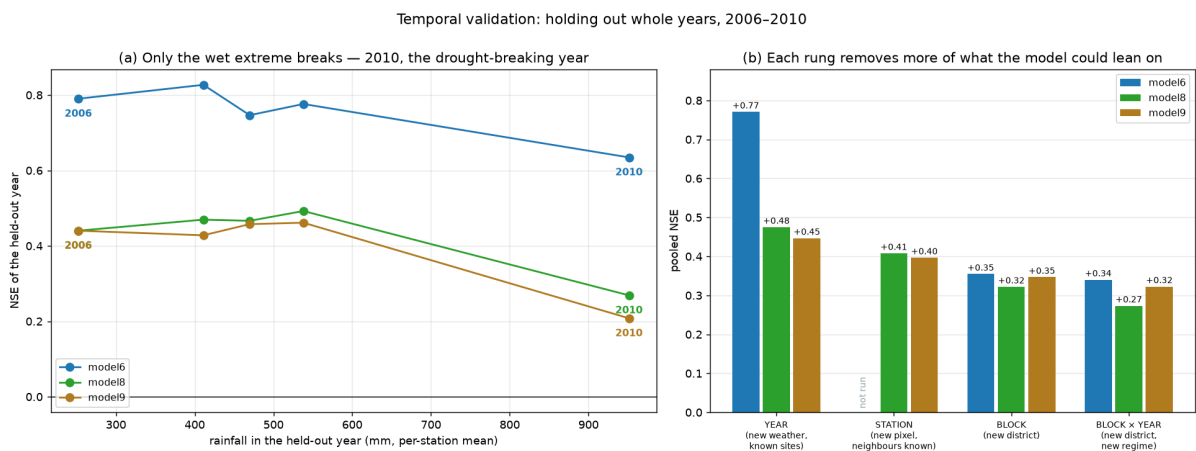


Figure 11: Temporal validation: leave-one-year-out across the drought break, and the block $\times$ year ladder.

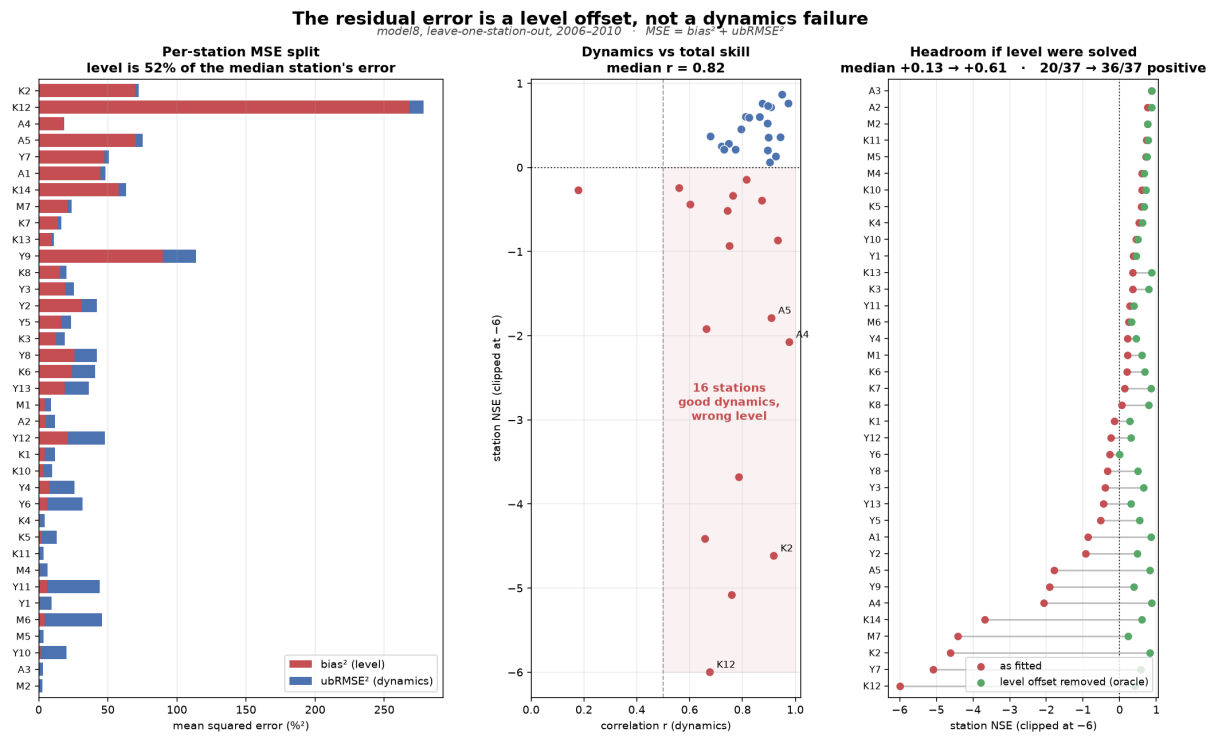


Figure 12: Error decomposition, $MSE = bias^2 + ubRMSE^2$, per station: a median 52 % of held-out MSE is attributable to level, while the dynamics are tracked at a median $r = 0.82$.

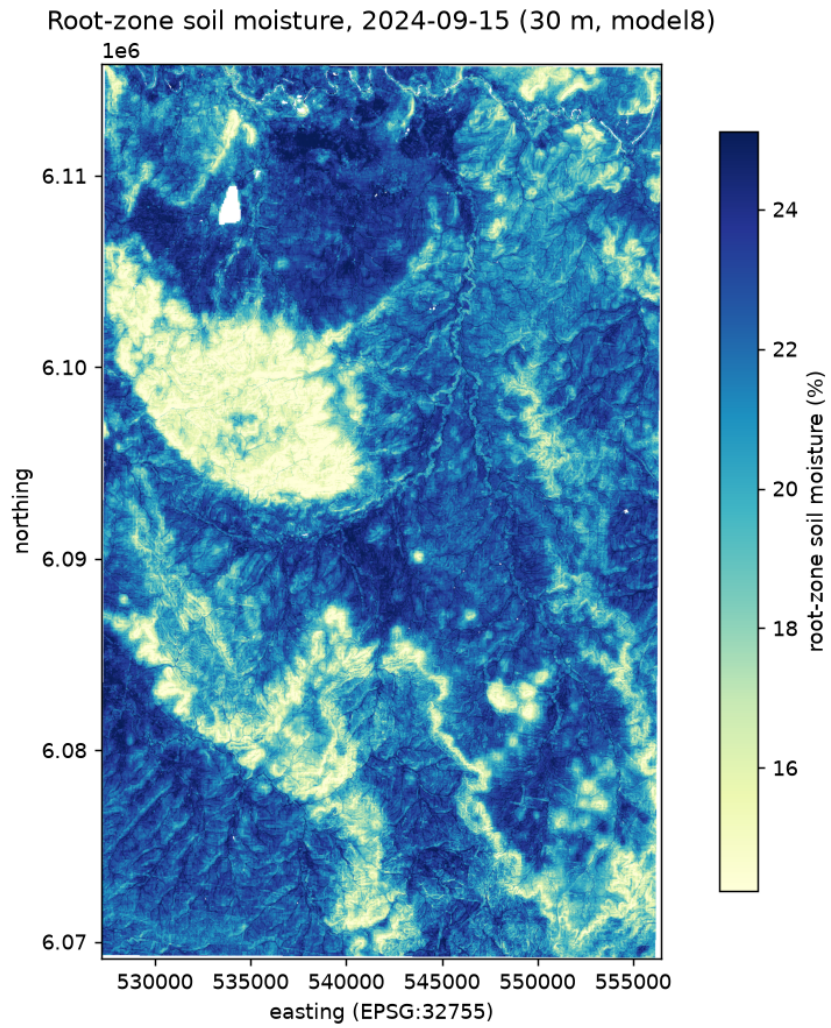


Figure 13: The process model as a product: 30 m field over Kyeamba for 2024-09-15, fourteen years after the training period and produced without SMIPS. Drainage lines and valley floors are wet; ridges and steep blocks are dry.

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